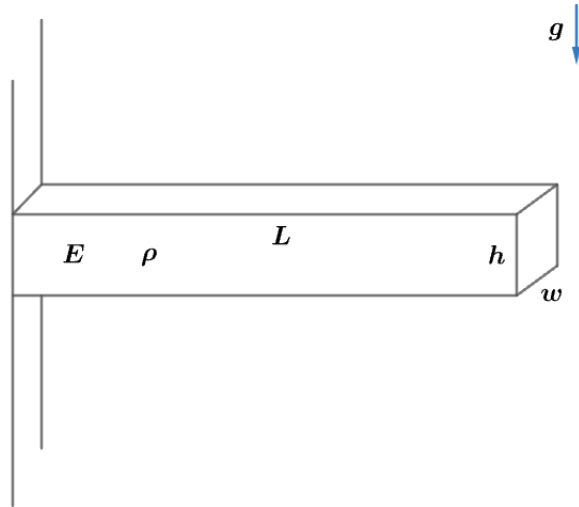


# 2008 F=ma Exam: Problem 20

Kevin S. Huang



Eliminate:

- (a)  $h \exp\left(\frac{\rho g L}{E}\right)$

If  $L = 0$ , the deflection should be zero.

- (b)  $2 \frac{\rho g h^2}{E}$

If  $L = 0$ , the deflection should be zero.

- (c)  $\sqrt{2Lh}$

Deflection should decrease as Young's modulus  $E$  increases.

- (e)  $\sqrt{3 \frac{EL}{\rho g h}}$

Deflection should decrease as Young's modulus  $E$  increases.

Thus, the answer is D.